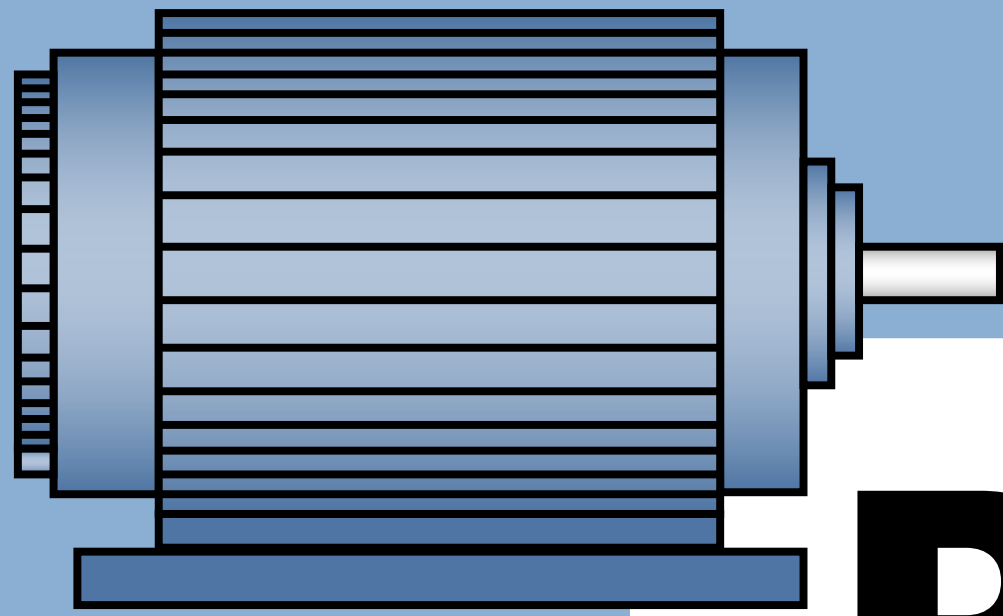




Group 13

PID Motor Controller

EN2111 - Electronic Circuit Design



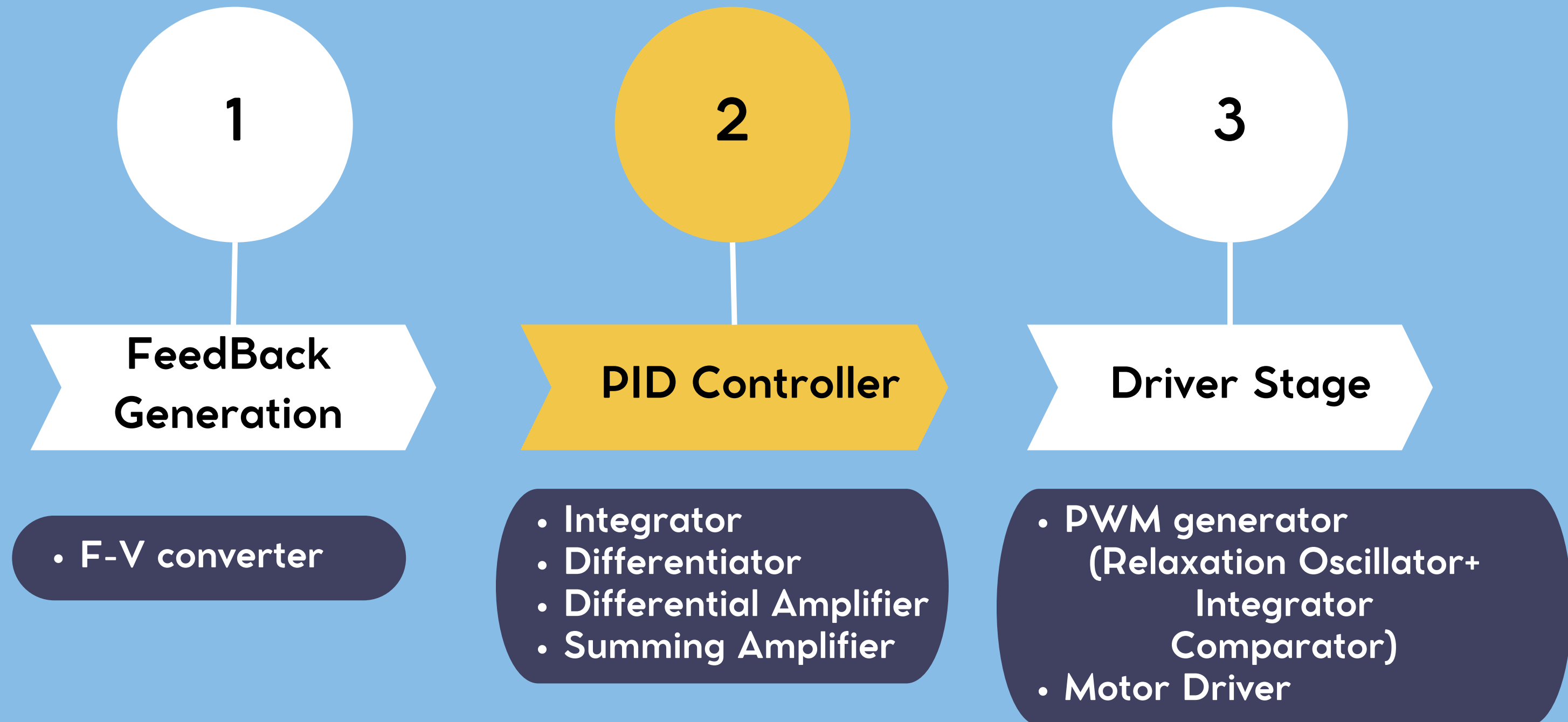
Introduction

In this project, we design an analog PID-based motor speed controller that maintains constant motor speed despite load variations. The system uses

- A reference speed input
- Feedback from motor encoder
- Analog PID control
- PWM-based motor drive

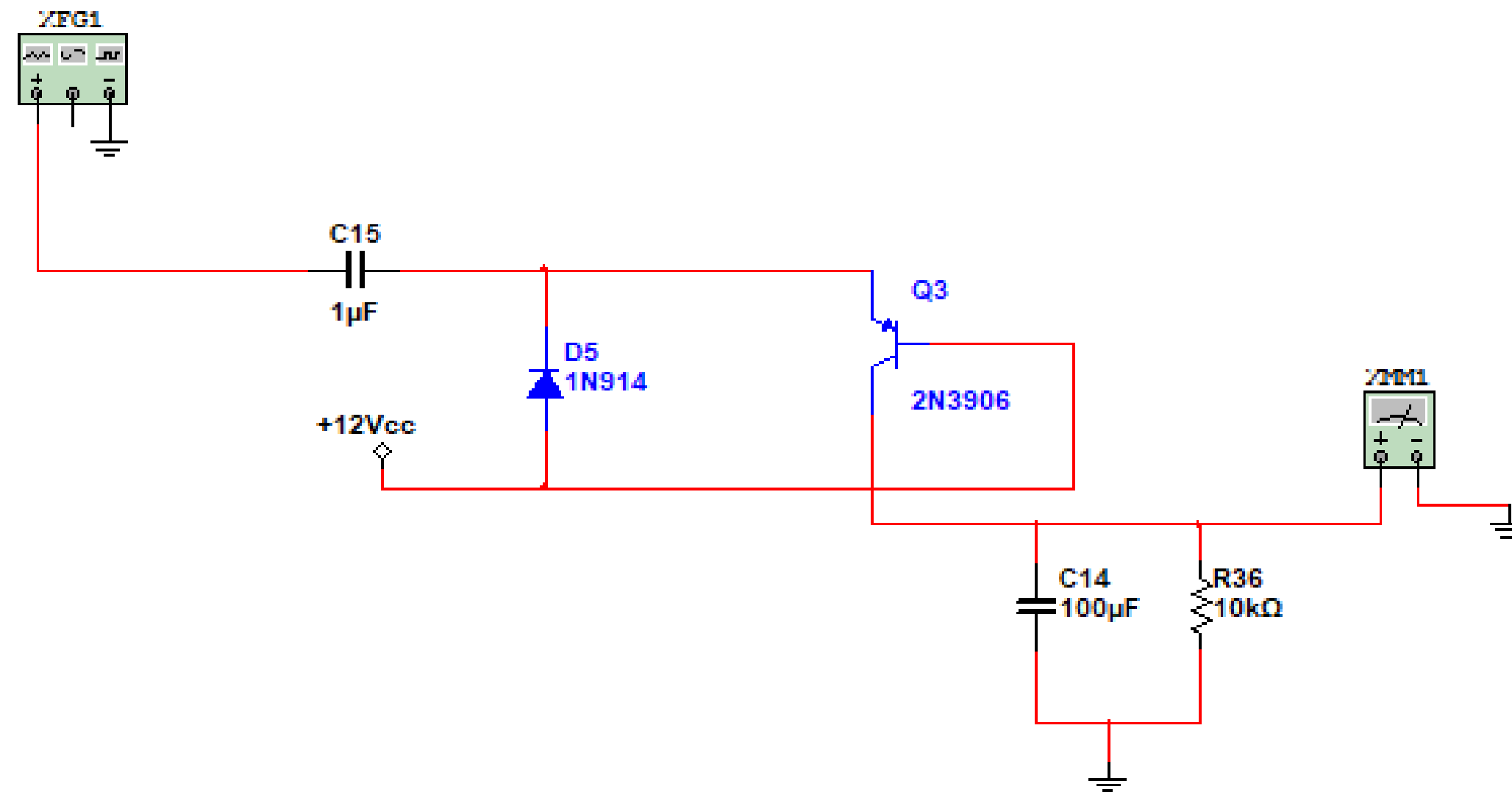


Block Diagram

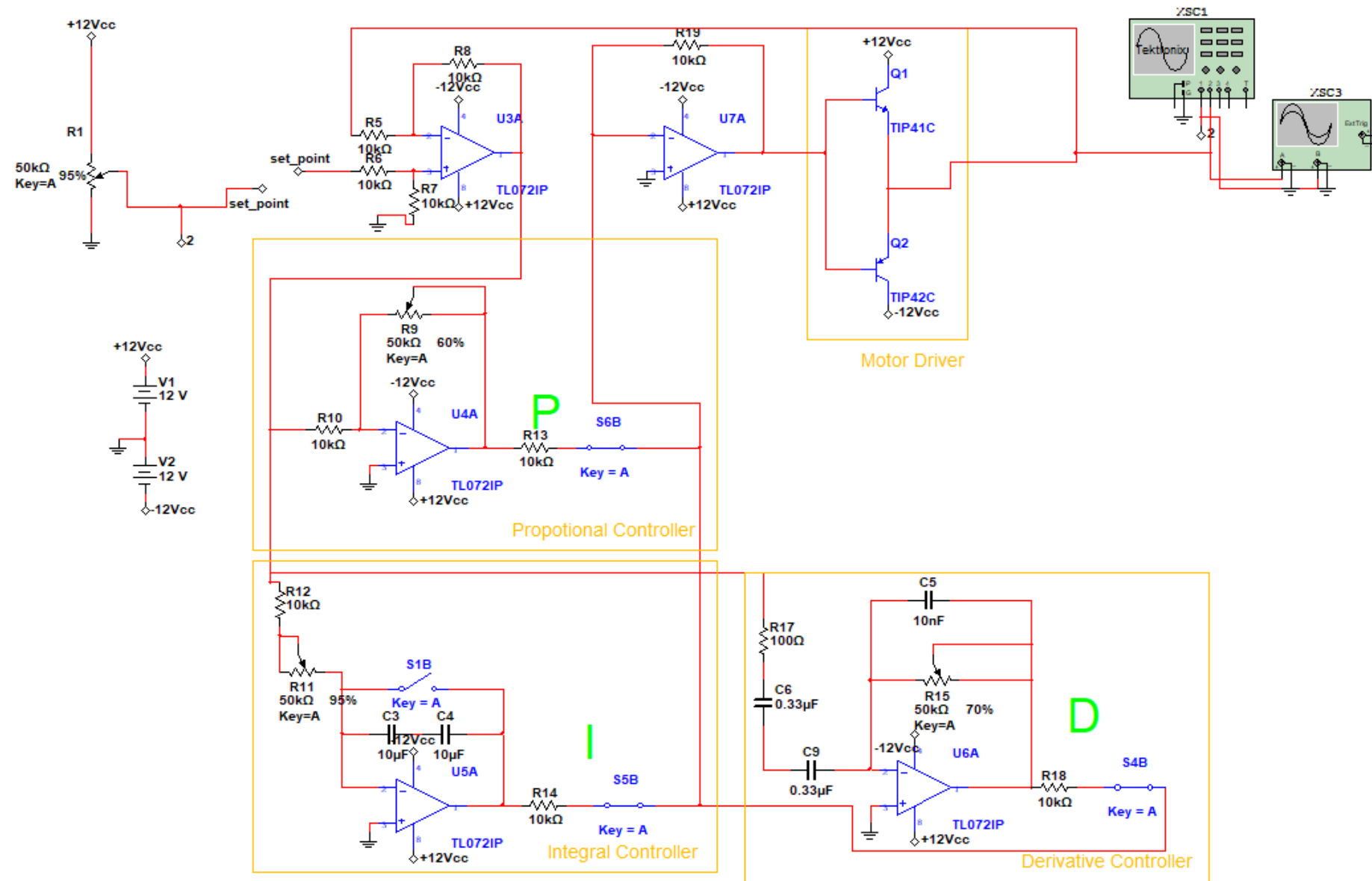


FeedBack Generation

F-V Conversion



PID Controller



This is the core of the system.

Reduce error between desired and actual speed

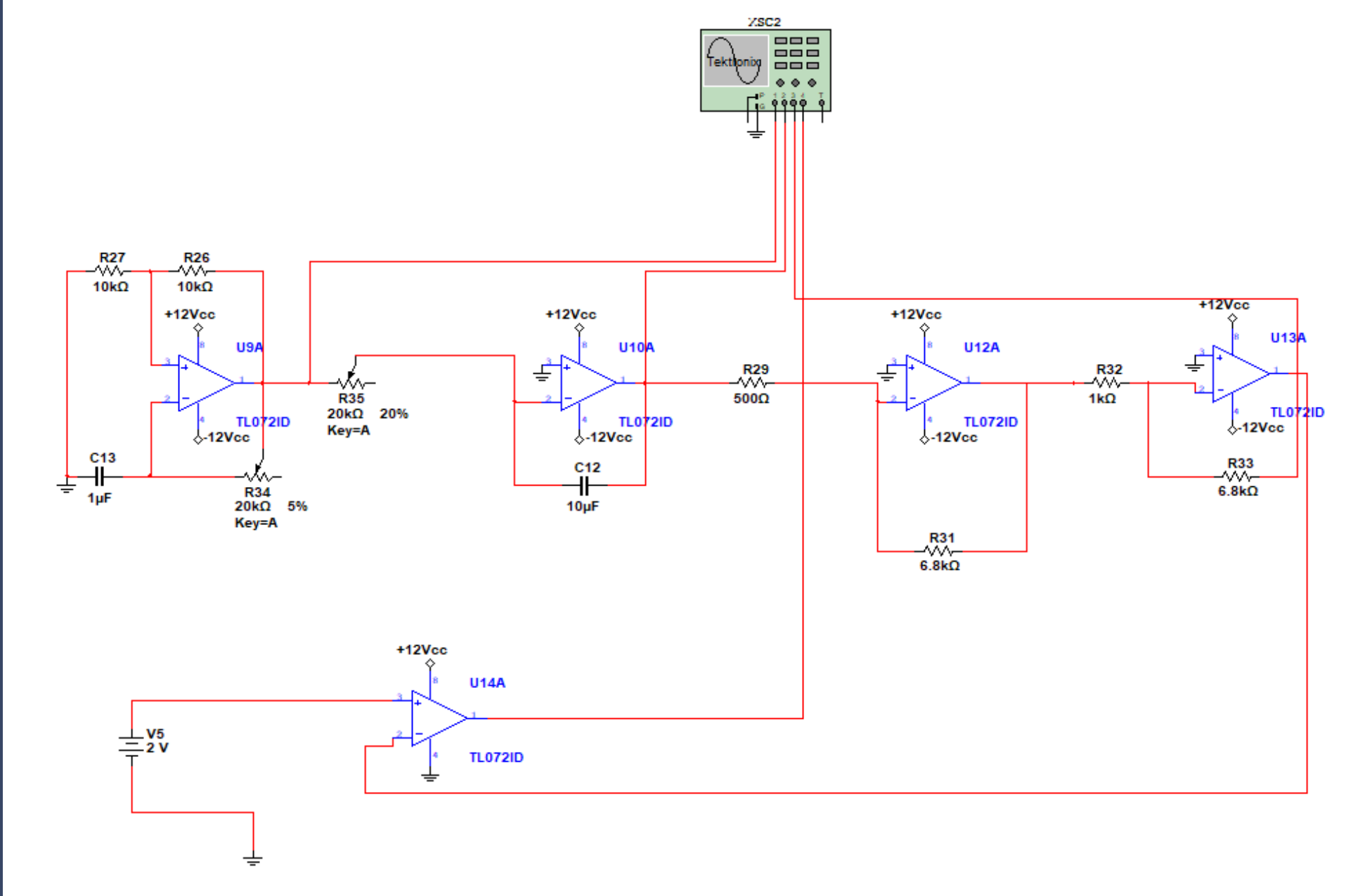
Improve stability and response

- Proportional (P): reacts to present error
- Integral (I): removes steady-state error
- Derivative (D): reduces overshoot and improves stability

Implemented using op-amp networks

$$u(t) = K_p e(t) + K_i \int_0^t e(t) dt + K_d \frac{de(t)}{dt}$$

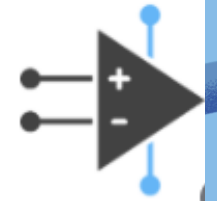
PWM Generation



Component Selection & Specification

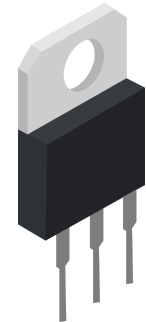
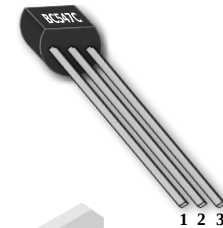
Op Amps-

- TL071 - Has higher slew rate, low noise
- LM741 - General purpose



Transistors-

- 2N3906P - switching/amplification
- TIP41C - Motor Driver



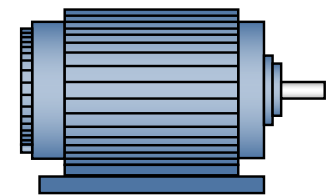
Diodes-

- 1N914 - Fast switching



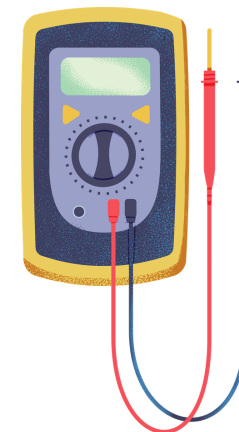
Motor -

- N20 - built in magnetic encoder

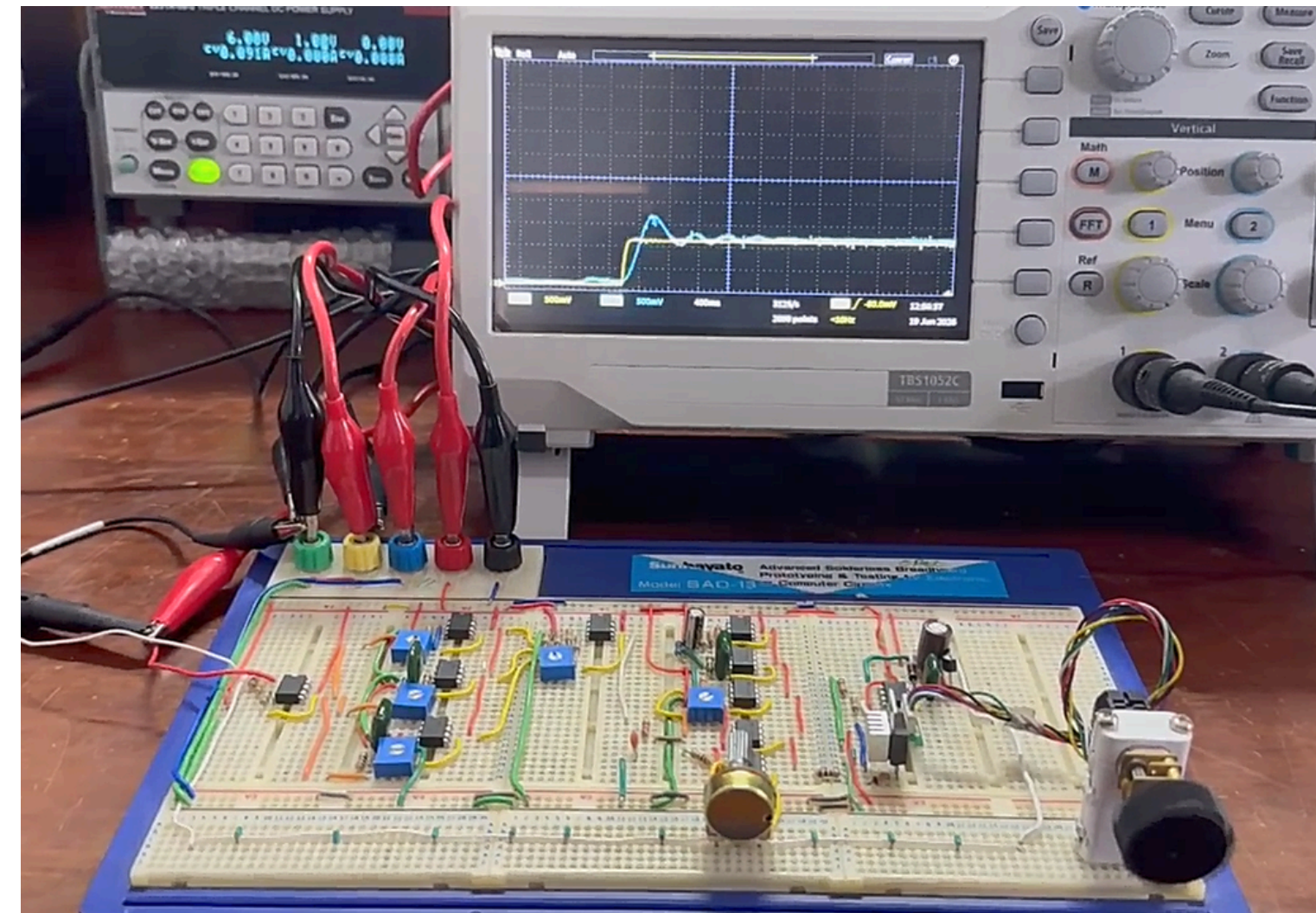
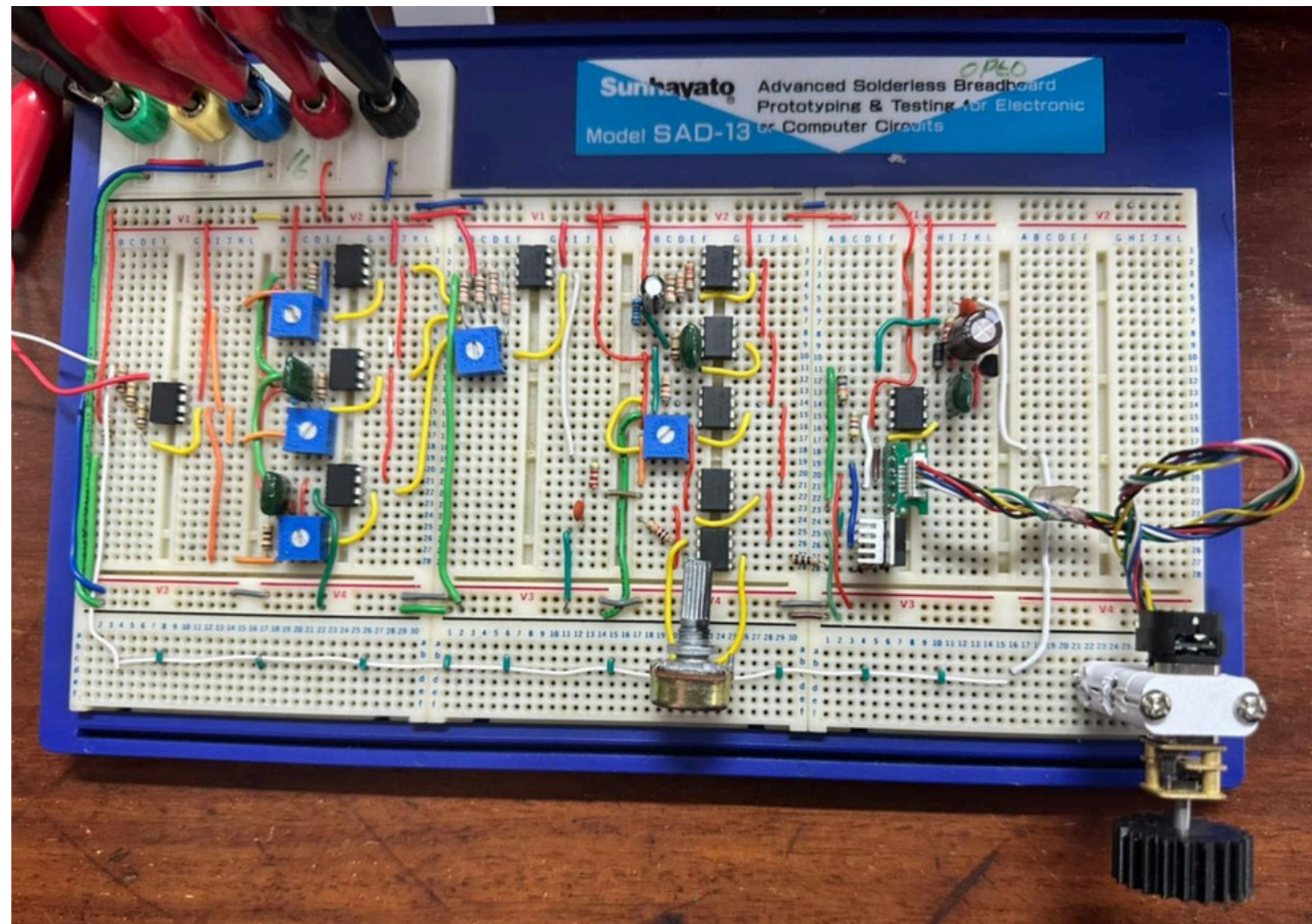


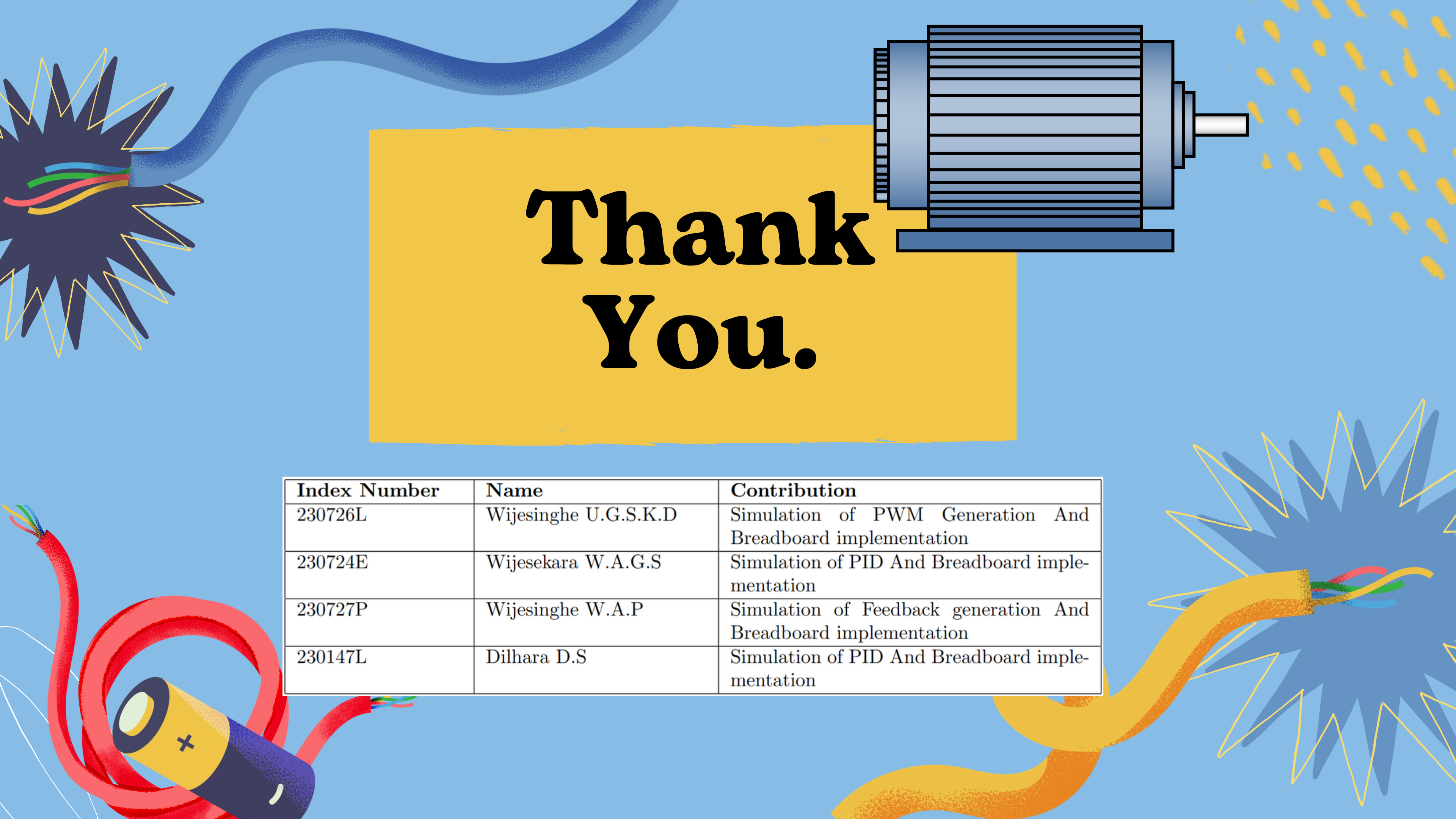
Supply Voltage-

- 6V
- $\pm 12V$ Dual Supply



Bread Board Implementation





Thank You.

Index Number	Name	Contribution
230726L	Wijesinghe U.G.S.K.D	Simulation of PWM Generation And Breadboard implementation
230724E	Wijesekara W.A.G.S	Simulation of PID And Breadboard implementation
230727P	Wijesinghe W.A.P	Simulation of Feedback generation And Breadboard implementation
230147L	Dilhara D.S	Simulation of PID And Breadboard implementation